Methods: frequency-resolved active-space downfolding of the electron–defect vertex and the defect spectral function

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We describe a first-principles route from a defect supercell to the electron–defect spectral function $A(\mathbf{k},\omega)$ of an isolated defect on an arbitrarily fine k-mesh. The scattering vertex is downfolded onto a small active manifold by an exact Feshbach partition, with the rest-space contribution resummed to all orders and resolved in frequency by a block Lanczos continued fraction. The dressed vertex is carried to fine k by a pair-Wannier transform, and the T-matrix is closed on a real-space cluster whose dimension is independent of the fine mesh. Each step is gated by a quantitative test: an operator-level identity, level-by-level comparison with a supercell reference at two supercell k-points, a leave-one-out test of the interpolation, and closure of the end-to-end chain onto the directly diagonalized quasiparticle levels to 0.4 meV.

I. OVERVIEW

The object of interest is the electron–defect self-energy of a dilute, randomly distributed defect,

$$\Sigma^{\mathrm{ed}}_{mn}(\mathbf{k},\omega) = n_d\, T_{mn}(\mathbf{k},\mathbf{k};\omega), \tag{1}$$

and the spectral function that follows from it,

$$A(\mathbf{k},\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\big[\omega + i\eta - H_0(\mathbf{k}) - \Sigma^{\mathrm{ed}}(\mathbf{k},\omega)\big]^{-1}. \tag{2}$$

Three obstacles stand between a supercell calculation and Eqs. (1)–(2). First, the scattering must be summed to all orders, not truncated at the Born level, because a defect that binds a state is not perturbative. Second, the sum must be resolved in frequency: freezing it at a reference energy is exact only there. Third, a supercell fixes both the defect concentration and the k-mesh, whereas Eq. (1) requires the isolated limit and a mesh dense enough to resolve $A(\mathbf{k},\omega)$.

The method addresses the three in sequence. Section II splits the Hilbert space and eliminates everything outside a small active manifold exactly. Section III constructs the bare vertex on the primitive grid and states the conditions under which that construction is exact. Section IV resums the eliminated space to all orders and to arbitrary ω. Section V carries the result to fine k, and Sec. VI closes the T-matrix in real space, where the concentration becomes a free parameter decoupled from the mesh. Section VII collects the tests.

Throughout, the illustrative system is a monolayer MoS_2 with a relaxed sulfur vacancy (V_S), and, where stated, the isovalent substitutions O_S and Se_S.

II. ACTIVE/REST PARTITION AND THE EXACT DOWNFOLDING IDENTITY

Let P_A project onto an active manifold A — n_b Bloch bands of the pristine crystal at each of N_k points of a coarse mesh, so $N_A = n_b N_k$ — and let $P_R = 1 - P_A$. With $H = H_0 + \Delta V$, where H_0 is the pristine Kohn–Sham Hamiltonian and ΔV the defect potential, Feshbach–Löwdin partitioning of $(\omega - H)^{-1}$ gives the projected Green function on A *exactly*,

$$G_A(\omega) = \big[\omega - \varepsilon_A - \widetilde{V}(\omega)\big]^{-1}, \tag{3}$$

with the dressed vertex

$$\widetilde{V}(\omega) = \frac{1}{N_k}\Big[M_{AA} + \Sigma^R(\omega)\Big], \qquad M_{ab} = \langle\psi_a|\Delta V|\psi_b\rangle, \tag{4}$$

and the rest self-energy

$$\Sigma^R(\omega) = V_{AR}\,\big[\omega - P_R H P_R\big]^{-1} V_{RA}. \tag{5}$$

Equation (5) contains *every* path that leaves A, scatters an arbitrary number of times within R, and returns — all orders in ΔV restricted to R, zeroth order in A. The complementary paths, those that re-enter A, are generated by inverting Eq. (3) or, equivalently, by the ladder of Sec. VI. Because $P_A P_R = 0$, no path is counted twice; this is the structural reason a T-matrix built on $\widetilde{V}$ with an active-space propagator requires no subtraction.

Two consequences are used later. (i) Since Σ^R depends on ω, the spectral weight carried by A is less than unity, with the quasiparticle renormalization $Z = [1 - \partial_\omega \widetilde{V}]^{-1}$; this must be reported when A-space spectral weights are compared with total ones. (ii) The factor $1/N_k$ in Eq. (4) is the Born–von-Kármán normalization: a calculation on an N_k-point mesh describes one defect per N_k primitive cells. N_k is therefore not only a sampling parameter but also the dilution, a point we return to in Sec. III C.

III. THE DEFECT VERTEX ON THE PRIMITIVE GRID

A. Folding the supercell potential

ΔV is obtained as the difference of the self-consistent local potentials of a defect supercell and the corresponding pristine supercell, aligned in the vacuum. It is represented on the supercell real-space grid. The matrix element between primitive-cell Bloch states requires it folded onto the primitive grid with the exact momentum transfer,

$$V_{\mathbf{q}}(\mathbf{r}) = \sum_{\mathbf{R}\in\text{cell}} \Delta V(\mathbf{r}+\mathbf{R})\, e^{i\mathbf{q}\cdot(\mathbf{r}+\mathbf{R})}, \qquad \mathbf{q} = \mathbf{k}' - \mathbf{k}, \tag{6}$$

where $\mathbf{R}$ runs over the primitive lattice vectors contained in the supercell box. Equation (6) is evaluated on the fly for each pair, with the phase running over the *whole* box and $\mathbf{q}$ taken exactly; wrapping $\mathbf{q}$ into the first Brillouin zone without the compensating phase would introduce a spurious factor. The implementation follows the convention of Ref. 9.

B. Commensurability of the fold

The sum in Eq. (6) is realized as a real-space modulo map, $r_i^{\text{cell}} \to r_i^{\text{cell}} \bmod n_i^{\text{prim}}$. This is exact only if

$$n_i^{\text{cell}} = N_i^{\text{sc}}\, n_i^{\text{prim}} \qquad \text{on every axis,} \tag{7}$$

i.e. the ratio must *be* the supercell multiplicity, not merely an integer. The failure is silent: in-gap states still appear at roughly the right energies while C_3-degenerate doublets split by tens of meV and $\|\widetilde{V}\|$ changes by tens of percent. A merely integral ratio fails just as badly — a 240-point cell of a 6× supercell folded onto a 30-point primitive grid (ratio 8) returned 15 states in a window that holds 5. Degeneracy splitting of a symmetric defect is the cheap fingerprint; on commensurate grids the C_3 pairs here are degenerate to 0–4 μeV. The code prints both grids and the detected multiplicity and aborts when Eq. (7) fails.

Equation (7) can be satisfied from either side. Rather than refine the primitive grid to match a fine potential grid, we band-limit the potential onto a coarser commensurate grid by truncation in $\mathbf{G}$-space, which is exact for every Fourier component the coarse grid supports. For the case studied, $240\times240\times300 \to 162\times162\times216$ discards 4×10^{-8} of the spectral weight of ΔV and pairs with the natural $27\times27\times216$ primitive grid; the cost of Eq. (6) falls in proportion to the grid, and quasiparticle levels move by ≤ 0.1 meV.

C. The k-mesh is free; N_k is the dilution

The supercell is a container that captures one isolated defect, not a periodicity we wish to impose. Because the phase in Eq. (6) runs over the whole box, $V_{\mathbf{q}}$ is the finite-box transform of a *localized* ΔV and is defined, and generically nonzero, for arbitrary $\mathbf{q}$. The k-mesh therefore need not match the supercell.

The distinction is quantitative. Treating the box as a periodicity would set $\langle\psi_{\mathbf{k}}|\Delta V|\psi_{\mathbf{k}'}\rangle = 0$ unless $\mathbf{k}-\mathbf{k}' \in \{\mathbf{G}_{\text{sc}}\}$. Measured on a 12×12 mesh against a 6×6 box, matrix elements between k-points differing by a non-supercell vector reach 0.975 of the largest within-supercell one: the periodic reading would discard most of the scattering channels.

Two inequivalent calculations therefore share the name "12×12":

(a) *Coset-decomposed.* Since ΔV of a strictly periodic array cannot connect them, the mesh splits into $(N_k/N^{\text{sc}})^2$ cosets of the supercell reciprocal lattice, each an independent run of the designed size, and each folding to one supercell k-point. The dilution stays at $1/N^{\text{sc}}$, which is what makes it comparable to a supercell calculation. This is a validation device (Sec. VII B), not the physics target.

(b) *Direct.* One run on the full mesh: one defect per N_k cells, with $\mathbf{q}$ resolved throughout the zone. This is the dilute limit the method targets.

The two agree where they must: the coset-diagonal blocks of the direct 144-k vertex reproduce the four shifted 36-k runs to 2×10^{-6} on a block scale of 66, compared gauge-invariantly through block eigenvalues.

IV. THE REST SELF-ENERGY BY BLOCK LANCZOS

A. Krylov construction

Equation (5) is a resolvent of the dressed rest operator $P_R H P_R$ between the N_A source vectors

$$|\chi_a\rangle = P_R\,\Delta V\,|\psi_a\rangle. \tag{8}$$

We block-tridiagonalize $P_R H P_R$ in the Krylov space generated by the block χ, with block width N_A,

$$P_R H P_R\, Q_j = Q_{j-1}B_{j-1}^{\dagger} + Q_j A_j + Q_{j+1} B_j, \qquad Q_0 R_0 = \text{qr}(\chi). \tag{9}$$

Full reorthogonalization against all stored blocks is used; the three-term recurrence alone loses orthogonality catastrophically and produces ghost eigenvalues. This is why all blocks Q_j must be retained, and it is the memory bottleneck of the method (Sec. VIII).

Each step requires two operator applications with very different costs. Applying H_0 is diagonal in $\mathbf{k}$: kinetic energy in $\mathbf{G}$-space, one pair of FFTs for the local potential, and a projector ZGEMM for the nonlocal part. Applying ΔV scatters, so every output channel couples to every source channel,

$$(\Delta V\chi)_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{k}'} V_{\mathbf{k}-\mathbf{k}'}(\mathbf{r})\, \chi_{\mathbf{k}'}(\mathbf{r}), \tag{10}$$

a triple loop over grid points, output channels and source channels. It dominates the step and sets the scaling of Sec. VIII.

B. Continued fraction

For any complex ω the resolvent follows from the block continued fraction, evaluated from the bottom of the chain,

$$S_j = \left[\omega - A_j - B_j^\dagger S_{j+1} B_j\right]^{-1}, \qquad \Sigma^R(\omega) = \frac{R_0^\dagger S_1(\omega) R_0}{N_k}. \tag{11}$$

The inverse is constructed, never expanded, so there is no convergence-radius condition: a dressed rest state that a Neumann ladder would fail to converge is carried exactly. One chain serves every ω and every broadening.

Two properties of Eq. (11) matter in practice. The poles of Σ^R are exactly the Ritz values of the block-tridiagonal Krylov Hamiltonian, so their number in any window can be counted by inertia (a block $LDL^\dagger$ sweep) without computing eigenvectors. And the residue of a pole is $R_0^\dagger|n\rangle\langle n|R_0/N_k$, which is rank one; a pole whose eigenvector has little weight on the source block contributes negligibly even if it lies inside the region of interest.

For the case studied, of the $N_A N_S = 25\,344$ poles, 25 lie inside the active manifold and four inside the fundamental gap — yet a dense scan shows $\|\Sigma^R\|$ rising smoothly and monotonically from 0.398 to 0.863 across the whole gap, so those four carry negligible residue. One pole near +2.65 eV does not: $\|\Sigma^R\|$ reaches 41.9 and the diagonal changes sign. It is a dressed rest state about 1 eV above the conduction edge, and it is exactly the kind of object that defeats a truncated ladder.

C. Frequency dependence in practice

Freezing Σ^R at a reference ω_0 is exact at ω_0 and degrades linearly away from it. In the case studied ω_0 lies below the gap, and freezing costs 18.2 meV on the a_1 in-gap level and 23.5 meV on the deeper doublet — larger than every other error in the chain. It is also unnecessary. Because all poles of Σ^R lie outside the active manifold except those identified above, Σ^R is analytic across the gap with the nearest singularity more than 3 eV away, and a low-order Chebyshev fit on a handful of nodes reproduces a held-out node to 0.026 meV and the quasiparticle fixed points to 0.201 meV.

When a wider window or a very dense ω grid is wanted, two remarks apply. The pair-Wannier transform of Sec. V is *linear*, so writing $\widetilde{V}(\omega) = \sum_i f_i(\omega)\, C_i$ places the entire frequency dependence on scalar coefficients: the C_i are transformed once and any ω costs a weighted sum. Alternatively the pole–residue form is exact and, because the residues are rank one, its transform factorizes into single-index vector transforms, $\mathcal{W}_n(\mathbf{R}_e, \mathbf{R}_p) = \tilde{u}_n(\mathbf{R}_e) \otimes \tilde{u}_n^*(\mathbf{R}_p)$. In either case the continued fraction, not the transform, is the quantity to economize.

V. WANNIER REPRESENTATION OF THE DRESSED VERTEX

A. Pair-Wannier transform

Because the defect breaks translational symmetry, $\widetilde{V}(\mathbf{k}, \mathbf{k}')$ is not a function of $\mathbf{k} - \mathbf{k}'$ and cannot be interpolated with a single index. We use the two-index construction of Ref. 9,

$$\mathcal{V}(\mathbf{R}_e, \mathbf{R}_p) = \frac{1}{N_k^2} \sum_{\mathbf{k}\mathbf{k}'} e^{+i\mathbf{k}\cdot\mathbf{R}_e} e^{-i\mathbf{k}'\cdot\mathbf{R}_p}\, U^\dagger(\mathbf{k})\, \widetilde{V}(\mathbf{k}, \mathbf{k}')\, U(\mathbf{k}'), \tag{12}$$

with $U(\mathbf{k})$ the Wannier gauge matrices. *Both* indices are electron Wannier lattice vectors: $\mathcal{V}(\mathbf{R}_e, \mathbf{R}_p) = \langle w_{\mathbf{R}_e}|\Delta V|w_{\mathbf{R}_p}\rangle$. The defect is not a third index; it is where ΔV lives, hence the point about which the kernel decays. (An equivalent parametrization transforms the electron index at fixed momentum transfer and the transfer separately; the two are related by the shear $\mathbf{R}_e' = \mathbf{R}_p - \mathbf{R}_e$.)

The construction requires an isolated composite band group so that U is unitary; disentanglement would make P_A mesh-dependent. For MoS_2 the 11-band $d + p$ manifold is such a group, giving $\|U^\dagger U - \mathbb{1}\| = 2 \times 10^{-10}$ and a gauge-invariant spread $\Omega_I = 17.70$ Å^2 with $\Omega_D = 0.030$ Å^2. Critically, the wannierization must be performed on the *same* wavefunctions used to build the vertex: independent diagonalizations differ by arbitrary per-state phases.

B. The origin convention

$\mathcal{V}$ is built on Born–von-Kármán representatives, and images differ by $N^{\rm sc}\mathbf{a}$. On the coarse mesh $e^{i\mathbf{k}\cdot N^{\rm sc}\mathbf{a}} = 1$, so the choice of representative is invisible — the transform inverts exactly whatever is chosen. On a *finer* mesh it is not unity, and a poor choice aliases. When the defect sits at the box centre rather than the origin, choosing representatives about the origin piles the kernel onto the cell boundary and turns a 1 meV prediction into a 300 meV one. The conventional remedy is the translation

$$\Delta V(\mathbf{r} + \mathbf{R}_{\rm def}) \iff M(\mathbf{k}, \mathbf{k}') \to e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{R}_{\rm def}} M(\mathbf{k}, \mathbf{k}'), \tag{13}$$

after which the ordinary origin-centred Wigner–Seitz construction applies; choosing representatives about $\mathbf{R}_{\rm def}$ is the same operation.

$\mathbf{R}_{\rm def}$ should be read from the structure — the atom present in the pristine cell and absent from the defect cell — rather than from the kernel maximum, which returns a representative valid only on the lattice it was found on: $(3, 3) \equiv (-3, -3)$ modulo 6 but not modulo 12. Note the asymmetry: *truncating* a coarse-mesh kernel is insensitive to the choice; only interpolation is.

C. Validation by leave-one-out

Locality of $\mathcal{V}$ in both arguments is the assumption the whole route rests on, so it is measured rather than argued. From a direct 144-k vertex in a single Wannier gauge we build the kernel from the Γ coset alone — a complete 6×6 dataset, 6.2% of the pairs — and predict the rest. The kernel falls from 1.46 on site to 2×10^{-4} at $6a$, and the 6×6 and 12×12 curves, the same quantity measured in two Born–von-Kármán cells, coincide wherever they overlap: the smaller cell truncates early rather than aliasing.

Element-wise in the same gauge, and quoted as $\delta M\,\mathrm{Ry}/N_k$ (the scale at which a vertex element enters H_{eff}), the 93.8% of pairs never seen by the construction are predicted with median 0.744 meV (p90 1.03, max 2.44); the training pairs return at 4×10^{-15}, certifying the transform rather than the physics. Repeating the test on the *dressed* vertex at seven frequencies spanning the gap gives 0.61 meV at the bottom of the gap — better than the bare vertex, since the extra decaying resolvent makes $\widetilde{V}$ more local — rising to 0.88 meV at the top.

The same test quantifies the cost of a smaller active manifold. A five-band model built from the highest valence and four lowest conduction bands (an isolated group, so no disentanglement) would make the vertex construction 2.2× and the reorthogonalization 4.8× cheaper, but interpolates 4.4× worse (median 3.30 meV, p90 8.62). This is not a gauge that needs more optimization: all projection choices give the same gauge-invariant $\Omega_I = 21.81$ Å^2 against 17.70 for eleven bands, i.e. 4.36 versus 1.61 Å^2 per Wannier function. Excluding the anion p bands that hybridize with the manifold costs localization that no minimization recovers.

VI. T-MATRIX AND SPECTRAL FUNCTION

A. Real-space cluster

With $G^0_A = (\omega - \varepsilon_A)^{-1}$ the free active-space propagator,

$$T(\omega) = \left[1-\widetilde{V}(\omega)G^0_A(\omega)\right]^{-1}\widetilde{V}(\omega), \qquad G_A = G^0_A + G^0_A T G^0_A. \tag{14}$$

Solving Eq. (14) directly on a fine mesh is prohibitive. In the pair-Wannier representation it becomes

$$\mathcal{T} = \mathcal{V} + \mathcal{V}\,\mathcal{G}^A(\omega)\,\mathcal{T}, \tag{15}$$

which closes on a truncated real-space cluster because $\mathcal{V}$ is short-ranged and, in the gap, $\mathcal{G}^A$ decays exponentially. The dimension of Eq. (15) is set by the cluster radius, not by the fine mesh — this is the step that escapes the N_k^3 wall of Sec. VIII. It is the Koster–Slater construction [3] with a first-principles, frequency-dependent vertex in place of a model potential.

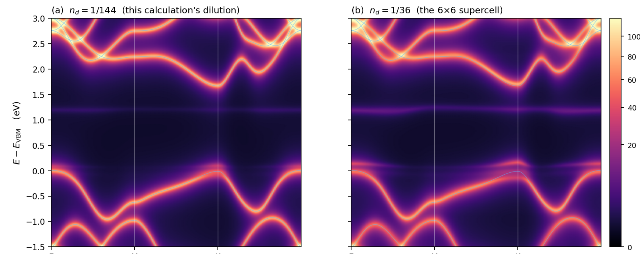

FIG. 1. Electron–defect spectral function $A(\mathbf{k},\omega)$ along Γ–M–K–Γ for the relaxed sulfur vacancy, at two concentrations, with the bare interpolated bands overlaid. The flat in-gap lines are the Koster–Slater bound states; their energy is $\mathbf{k}$-independent by construction while their weight is not.

B. Host Green function

The fine mesh enters only through a Brillouin-zone sum,

$$\mathcal{G}^A(\delta\mathbf{R};\omega) = \frac{1}{N_f^2}\sum_{\mathbf{k}_f} e^{i\mathbf{k}_f\cdot\delta\mathbf{R}}\left[(\omega+i\eta) - H^W(\mathbf{k}_f)\right]^{-1}, \tag{16}$$

with $H^W(\mathbf{k}_f)$ Wannier-interpolated from the same wannierization that supplies $U(\mathbf{k})$ in Eq. (12). Using the two objects from a single wannierization is a hard requirement: $\mathcal{V}$ and $\mathcal{G}^A$ must be expressed in the same orbital basis for their product to mean anything. As a consistency check, the interpolated host reproduces the valence-band maximum and the full active-manifold energy range of the direct calculation to all printed digits.

C. Self-energy and spectral function

Transforming back,

$$T_{mn}(\mathbf{k},\mathbf{k}';\omega) = \sum_{\mathbf{R}_e\mathbf{R}_p} e^{-i\mathbf{k}\cdot\mathbf{R}_e} e^{+i\mathbf{k}'\cdot\mathbf{R}_p}\,\mathcal{T}_{mn}(\mathbf{R}_e,\mathbf{R}_p;\omega), \tag{17}$$

and Eqs. (1)–(2) follow. Two remarks. First, the diagonal $T(\mathbf{k},\mathbf{k})$ gives the spectral function, whereas transport requires the off-diagonal $\sum_{\mathbf{k}'}|T(\mathbf{k},\mathbf{k}')|^2\delta(\varepsilon-\varepsilon_{\mathbf{k}'})$; both come from the same $\mathcal{T}$. Second, and central to the motivation, n_d in Eq. (1) is now a free physical parameter: the real-space cluster solves a genuine single-defect problem, so the concentration is no longer tied to the mesh as it is in Eq. (4).

In the gap $\mathcal{G}^A$ is real, so bound states are the roots of $\det[1-\mathcal{V}\mathcal{G}^A(\omega)] = 0$ — a determinant containing no $\mathbf{k}$. Their energies are therefore $\mathbf{k}$-independent while their spectral weight varies along the path, the latter being the overlap of the localized state with each Bloch state.

VII. VALIDATION

Each stage is gated separately, so that a failure is localized rather than absorbed into a final discrepancy. Table I collects the results.

A. Operator level

The application of ΔV inside the Lanczos recursion, Eq. (10), is compared against the independently constructed vertex matrix by projecting the operator applied to known band states onto the band basis. The two paths share no code beyond the potential file. Agreement is 2×10^{-11} relative on the 12×12 mesh and 2×10^{-14} on 6×6; the test is run at the start of every production chain and is the gate for any change to the fold.

B. Against a supercell reference

The coset construction of Sec. III C makes a supercell comparison possible at more than one supercell k-point while holding the concentration fixed. The three M cosets are symmetry-equivalent and the downfolding does not know it: run separately they return quasiparticle levels agreeing to every printed digit, matching the exact degeneracy of the supercell reference. A dedicated 107-atom calculation at the three M points supplies a reference no part of the downfolding has seen.

Levels agree with a rigid offset: $+25.5 \pm 1.7$ meV at Γ (5/5 features) and $+25.0 \pm 0.8$ meV at M (5/5). The two offsets agree to 0.5 meV — the constant is a property of ΔV, not of $\mathbf{k}$. It is a potential-reference convention: ΔV is vacuum-aligned while the reference is anchored on deep levels, and the difference is predicted with no fitting from two independently measured quantities (+17.9 meV here), leaving residuals of $+7.6 \pm 1.7$ and $+7.1 \pm 0.8$ meV.

The same machinery bounds what is worth chasing. Running the finer mesh directly rather than by cosets dilutes the defect four-fold with no change of alignment (the same ΔV, so the reference constant cancels exactly, and both meshes contain the same high-symmetry points so the pristine edges are identical). The true in-gap levels move by -11.1 and $+3.7$ meV: the supercell reference itself carries 4–11 meV of periodic-image error, of the same size as the residual above.

C. End to end

Finally the assembled chain — Wannier gauge, pair kernel, cluster truncation, fine-mesh $\mathcal{G}^A$, Dyson solve — is closed onto the coarse-mesh result. The deep in-gap bound state, a root of a determinant with no $\mathbf{k}$ in it, comes out at +1.2019 eV against +1.2023 eV from direct diagonalization of $H_{\text{eff}}(\omega)$: 0.38 meV. A systematic error anywhere in the chain could not survive this comparison.

TABLE I. Validation summary. Energies in meV unless noted.

Stage	Test	Result
Fold	C_3 degeneracy on commensurate grid	0–4 μeV
Vertex	operator identity, 12×12	2×10^{-11}
Vertex	operator identity, 6×6	2×10^{-14}
Vertex	direct vs. coset-diagonal blocks	2×10^{-6}
Chain	N_S convergence ($16 \to 24$)	0.8
Downfold	vs. supercell at Γ (5/5)	$+7.6 \pm 1.7$
Downfold	vs. supercell at M (5/5)	$+7.1 \pm 0.8$
Reference	periodic-image error of the reference	4–11
ω fit	held-out node	0.026
ω fit	quasiparticle fixed points	0.201
Interpolation	leave-one-out, bare vertex (median)	0.744
Interpolation	leave-one-out, dressed vertex	0.61–0.88
End to end	cluster vs. direct, deep level	0.38

A second in-gap feature 68 meV above the valence edge differs by 7.3 meV, and doubling the frequency resolution moves it by only 1.5 meV, so it is not a grid artefact. At $\eta = 50$ meV that state is hybridized with the valence continuum: it is a resonance, and the peak of $|T|$ is not the same quantity as a quasiparticle fixed point computed on a discrete coarse spectrum that has no continuum to hybridize with.

VIII. COMPUTATIONAL PARAMETERS, SCALING AND COST

A. Parameters

Ground state and pristine band structure with Quantum ESPRESSO [10, 11], PBE [13], $E_{\text{cut}}^{\text{wfc}} = 50$ Ry, $E_{\text{cut}}^{\rho} = 200$ Ry, a 2D truncated Coulomb treatment, and a 24 Å vacuum. The defect supercell is 6×6 (107–108 atoms), relaxed. The pristine calculation supplying the active manifold uses a $27 \times 27 \times 216$ FFT grid, 20 bands and a 12×12 k-mesh; the potential is band-limited to $162 \times 162 \times 216$ per Sec. III. The active manifold is the 11-band $d + p$ group, $N_A = 1584$. Chains use $N_S = 16$ blocks at 12×12 and $N_S = 24$ at 6×6. Wannier functions from Wannier90 [12], 11 per cell, Mo-d and S-p projections, no disentanglement. The T-matrix uses a cluster radius $4a$ (61 cells, dimension 671), a 96×96 mesh for Eq. (16), and $\eta = 50$ meV.

Note that a host calculation must declare every atomic species appearing in any defect cell, including species absent from the pristine cell, since the nonlocal projector count is indexed by the *cell's* species list.

B. Scaling

At fixed rank count the vertex application, Eq. (10), costs

$$t_{\text{fold}} \propto \frac{N_r\, n_b\, N_k^3}{n_{\text{pool}}}, \tag{18}$$

an N_k^2 channel double loop times N_k columns per rank; full reorthogonalization carries the same N_k^3 with an additional N_S^2. Measured, $36 \to 144$ k-points take the multiply from 4.1 to 261 s, i.e. $64\times = 4^3$, confirming the law. Memory scales as N_k^2, which is what limits the chain length. Reorthogonalization overtakes the vertex application around the ninth block and is the wall beyond this size; selective reorthogonalization, or the pole–residue representation of Sec. IV, are the levers.

C. Cost

On one 128-core node: the 144-k chain takes 140 min (16 blocks, 261 s per block in the vertex application). Post-processing is dominated by the continued fraction, 6 s per frequency at $N_A = 1584$, $N_S = 16$; the cluster solve and the pair-Wannier transform are 0.5 s and 0.1 s per frequency. Frequencies are independent, and an interleaved eight-worker split delivers 450 frequencies in 6 min 25 s (0.85 s per frequency, a 13.9× speedup). The complete spectral function of Fig. 1 — 450 frequencies, 181 k-points — therefore costs minutes once the chain exists, and any defect concentration afterwards is free because T is independent of n_d.

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